

Model

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1 Motivation

The model formalizes the dynamic problem of issuing warnings through LLMs, with the following key ideas

- **Saliency:** a warning makes reputation concerns more salient (what I show in Guriev et al 2025). It increases the weight you put on reputation.
- **Habituation:** repeated warnings decrease this saliency effect.
- **Loss of faith:** a false warning, when the news turned out to be true, decreases trust in the tool

Empirically, many of these dimensions have not been measured

- Saliency: effect when the warning is issued by LLM rather than a human, emphasising that the LLM just evaluates a risk and does not establish absolute truth, has not been measured
- Habituation: need to look what exists on the literature, but my sense is not much
- Loss of faith: same

2 Model

2.1 Players and Timing

Individuals are positioned in a line (simple network structure for the moment). Individual i is in the i th position in this line.

We consider the infinite horizon game where, in each period t , the first player in the line receives a piece of information. Each piece of information is characterized by two dimensions: its veracity true or false $v_t \in \{0, 1\}$, where $v_t = 1$ if the news is true (drawn according to a Bernoulli distribution with parameter $p = P[v_t = 1]$, independent across periods) and whether it is appealing to individual i ξ_{it} (iid drawn from F that does not depend on v_t). We assume F is a generalized type I extreme value distribution.

After receiving the information, each individual i receives simultaneously up to two signal

- With probability λ_t (i.i.d drawn for each individual in each period) a perfectly informative signal about the truth (coming for instance from trustworthy sources). An informed individual is denoted I_t . You can be informed in one period and not in the next.
- A warning $a_t \in \{0, 1\}$, generated by the LLM as described below.

Upon receiving these signals, individual i decides whether to share the news. Finally at the end of the period, everyone is informed on the state.

We might examine later the case where warnings are individualized, but for the moment we assume if a warning is triggered it reaches everyone

Warnings. The LLM can issue informative (useful) warnings or not $U \in \{0, 1\}$. If it is useful $U = 1$ (probability r , prior common to all agents), it produces a signal s_t that the news is false. We assume there exist two distributions G_0, G_1 with common support, such that:

$$s_t \sim \begin{cases} G_0(\cdot) \text{ if } v_t = 1 \text{ (true),} \\ G_1(\cdot) \text{ if } v_t = 0 \text{ (false),} \end{cases} \quad \text{when } U = 1,$$

The distributions are such that the likelihood ratio $\Lambda(s) = \frac{g_1(s)}{g_0(s)}$ is strictly increasing in s . A higher signal indicates that the news is more likely to be false. The designer thus chooses the certainty cutoff s^* such that a warning is triggered ($a_t = 1$) if $s_t \geq s^*$. If it is not useful, it produces a warning unrelated to the state with probability $\Pr(a_t = 1 \mid U = 0) = \tau$.

Objective of the designer. The designer wants to promote circulation of true news and restrain circulation of fake news. Each exposure to true news generates benefit b and to false news loss l . The designer thus wants to maximize

$$\max_{s^*} \sum_t \beta^t [p b C_t(v_t = 1) - (1 - p) l C_t(v_t = 0)]$$

with $C_t(v_t = 1) = \sum_n \nu_t^n = \frac{1}{1 - \nu_t}$, where $\nu_t(v_t, a_t)$ is the probability that a random individual in the line shares. Given that they make no inferences based on previous sharing decisions, all individuals are identical.

Sharing decision. Normalize the utility of not sharing to 0. If user i shares, her utility is

$$V_{it} = \xi_{it} - \alpha_t R_t$$

where R_t is the belief that the receiver (individual $i + 1$) assigns to the sender (individual i) being informed, i.e is of type I_t , i.e having received the perfectly informative signal. This inference is made based on the sharing decision. The strength of this reputation concern is given by α_{it} .

Thus the sharing rule is a cutoff in ξ : $\xi_{it} \geq \alpha_t R_t$.

In addition, we assume that informed users do not share false news. We also make the assumption that, when deciding to share, individual i only takes into account her own information and does not make inferences based on the sharing decision of $i - 1$.

Salience decreases with habituation in the following way:

$$\alpha_{it} = \underbrace{\bar{\alpha}}_{\text{baseline}} + \underbrace{s_t}_{\text{salience stock}}, \quad s_t = (1 - \delta_s)s_{t-1} + \kappa a_t,$$

Updating of beliefs. At the end of the period, the state is revealed and all users are going to update their beliefs on the quality of the LLM (useful or not). We assume that they don't infer anything when the warning is not issued (no saliency). That implies updating happens only when a warning is issued.

Suppose $v_t = 0$

$$\begin{aligned} r_{t+1} &= P[U = 1 | a_t = 1] \\ &= \frac{P[a_t = 1 | U = 1]P[U = 1]}{P[a_t = 1]} \\ &= \frac{r_t(1 - G_1(s^*))}{r_t(1 - G_1(s^*)) + (1 - r_t)\tau} \end{aligned}$$

Suppose $v_t = 1$

$$\begin{aligned} r_{t+1} &= P[U = 1 | a_t = 1] \\ &= \frac{P[a_t = 1 | U = 1]P[U = 1]}{P[a_t = 1]} \\ &= \frac{r_t G_0(s^*)}{r_t G_0(s^*) + (1 - r_t)\tau} \end{aligned}$$